

行星雷达探测中的电动力学原理 从Maxwell方程组到火星南极地下水

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提 纲

电磁波在介质中的传播

雷达的表层探测

雷达的次表层探测

火星南极地下水的发现

电磁波在介质中的传播 | 方程推导

线性、各向同性、均匀的无源有耗介质

$$\nabla \cdot \mathbf{E} = 0, \quad \text{无源}$$

$$\nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \times \mathbf{B} = \mu \mathbf{J} + \mu \epsilon \frac{\partial \mathbf{E}}{\partial t}.$$

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial(\nabla \times \mathbf{B})}{\partial t}$$

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu \frac{\partial \mathbf{J}}{\partial t} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \mathbf{J} = \sigma \mathbf{E}$$

$$\nabla^2 \mathbf{E} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu \sigma \frac{\partial \mathbf{E}}{\partial t} = 0$$

电场的波动方程

电磁波在介质中的传播 | 方程求解

$$\nabla^2 \mathbf{E} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu\sigma \frac{\partial \mathbf{E}}{\partial t} = 0$$

考虑：沿+z方向传播的单色平面波

$$\mathbf{E}(z, t) = \mathbf{E}_0 e^{i(kz - \omega t)}$$

化简得 $k^2 = \mu\omega^2\epsilon + i\mu\omega\sigma$

设 $k = \beta + i\alpha$

可得

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{-1 + \sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}},$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{1 + \sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}}.$$

考虑：介电弛豫等非瞬时响应

复介电常数 $\epsilon^* = \epsilon + i\epsilon'$

改写为 $k^2 = \underbrace{\mu\omega^2\epsilon}_{\text{储存}} + \underbrace{i\mu\omega^2\left(\frac{\sigma}{\omega} + \epsilon'\right)}_{\substack{\text{耗散} \\ \text{极性分子弛豫} \\ \text{自由电荷传导}}}$

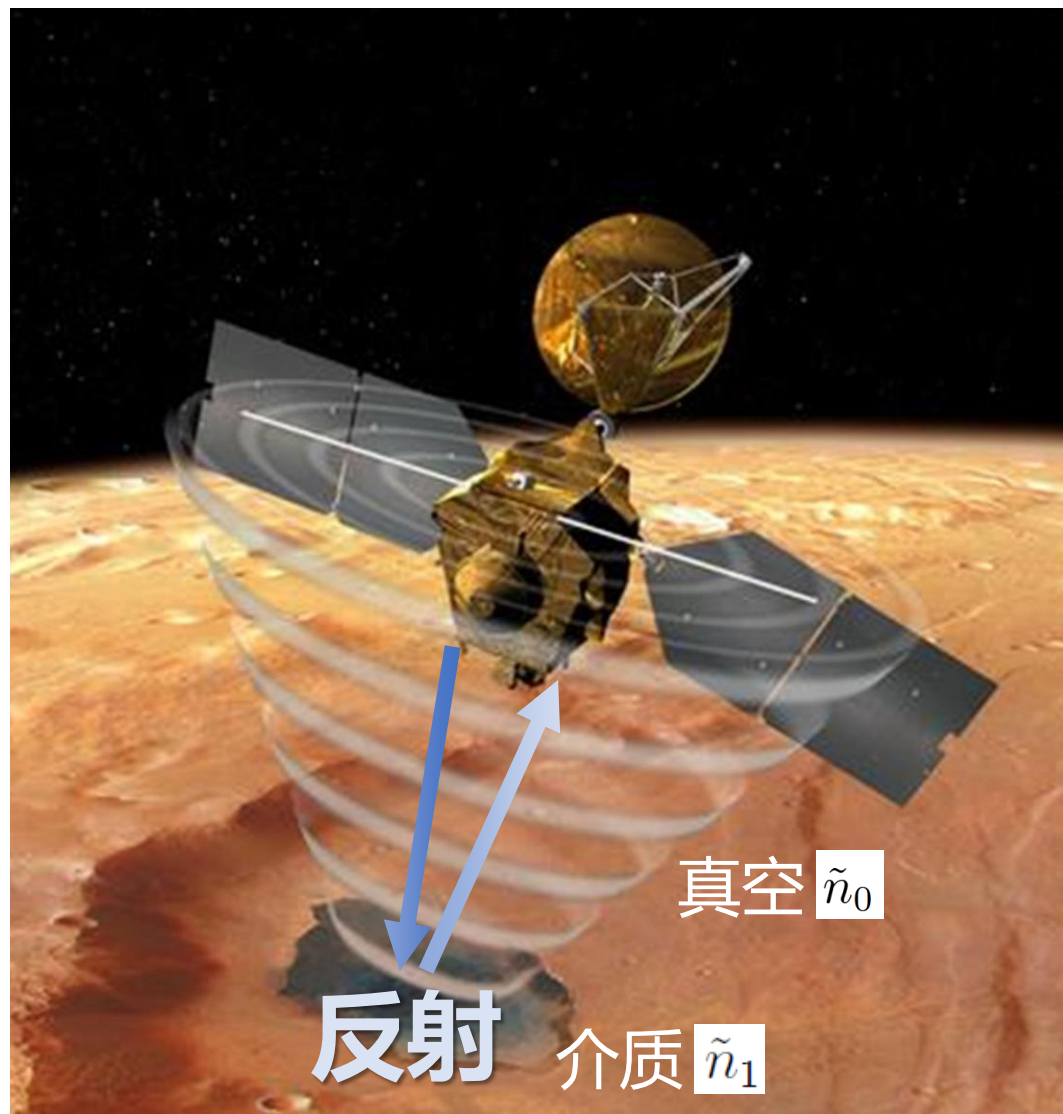
简记 $\epsilon^* = \epsilon + i\left(\frac{\sigma}{\omega} + \epsilon'\right)$

得 $k^2 = \mu\omega^2\epsilon^*$

同时 $\alpha = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{-1 + \sqrt{1 + \left(\frac{\epsilon''}{\epsilon}\right)^2}},$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{1 + \sqrt{1 + \left(\frac{\epsilon''}{\epsilon}\right)^2}}.$$

雷达的表层探测



影响

介质——**介电性质**——电磁波的反射

反演

$$\epsilon^* = \epsilon + i\left(\frac{\sigma}{\omega} + \epsilon'\right)$$

$$\tilde{n} = \sqrt{\frac{\mu\epsilon^*}{\mu_0\epsilon_0}} = \sqrt{\frac{\epsilon^*}{\epsilon_0}} = \boxed{n} + \boxed{i\kappa}$$

储存 耗散

非极化介质 $\mu = \mu_0$

菲涅尔定律
反射系数

$$r_{01} = \frac{\tilde{n}_0 - \tilde{n}_1}{\tilde{n}_0 + \tilde{n}_1}$$

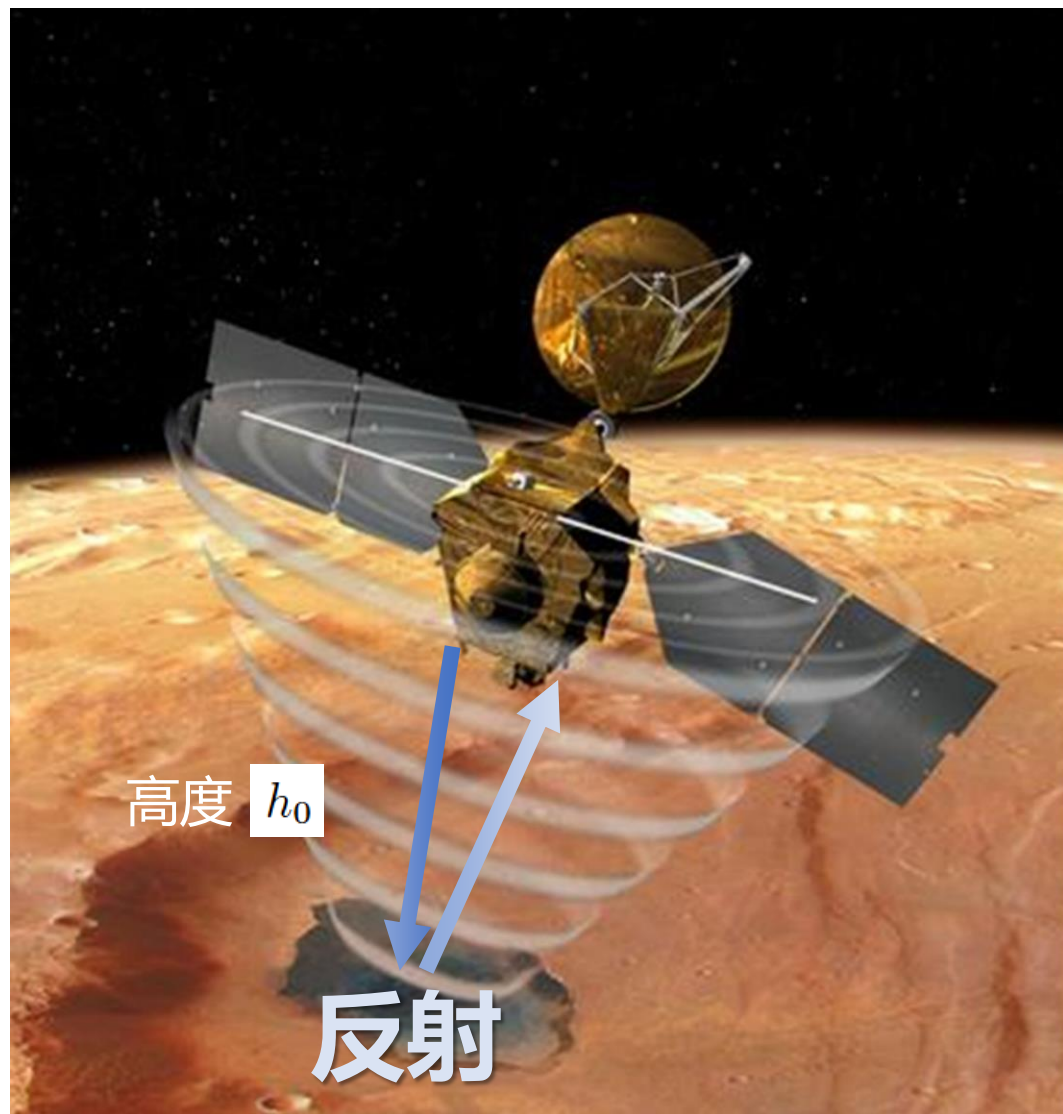
真空

$$\tilde{n}_0 = 1$$

功率反射率

$$R_{01} = |r_{01}|^2 = \left| \frac{1 - (n_1 + i\kappa_1)}{1 + (n_1 + i\kappa_1)} \right|^2 = \frac{(1 - n_1)^2 + \kappa_1^2}{(1 + n_1)^2 + \kappa_1^2}$$

雷达的表层探测



功率反射率 $R_{01} = |r_{01}|^2 = \left| \frac{1 - (n_1 + i\kappa_1)}{1 + (n_1 + i\kappa_1)} \right|^2 = \frac{(1 - n_1)^2 + \kappa_1^2}{(1 + n_1)^2 + \kappa_1^2}$

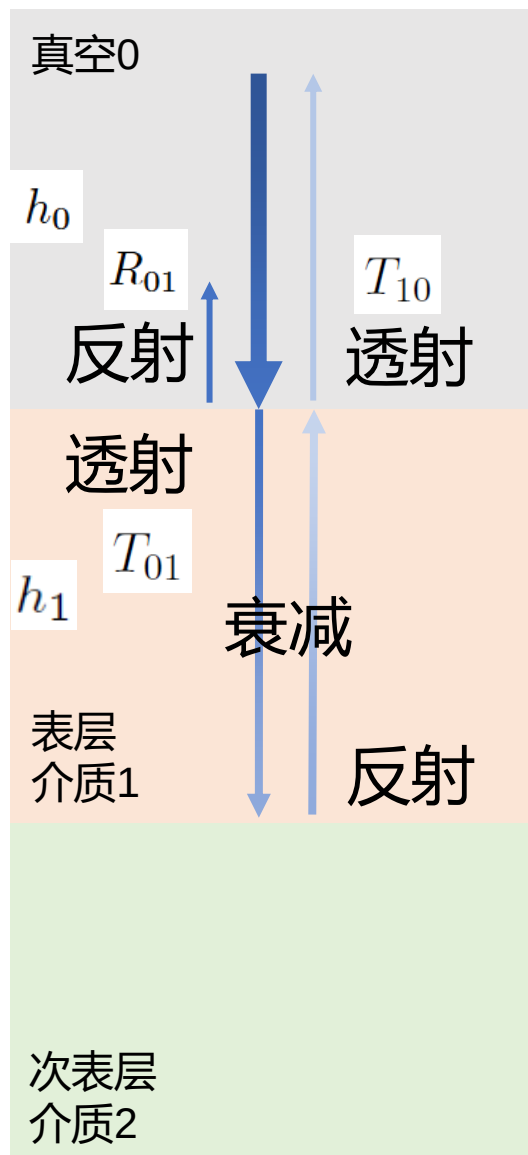
假定小耗散 $\kappa_1 \ll n_1$ $R_{01} \approx \left(\frac{1 - n_1}{1 + n_1} \right)^2$
耗散 储存

Friis传输方程 $P_s = P_t \frac{G^2 \lambda^2}{(4\pi)^2 (2h_0)^2} R_{01}$
接收功率 ↑
发射功率 ↓
天线增益 波长
高度

$$P_s = \boxed{\frac{P_t G^2 \lambda^2}{64\pi^2 h_0^2}} \left(\frac{1 - n_1}{1 + n_1} \right)^2$$

测量 已知 待求

雷达的次表层探测



表层雷达回波, 可知 $R_{01} \approx \left(\frac{1-n_1}{1+n_1}\right)^2$ $\tilde{n}_1$ 已知

透射 $T_{01} = T_{10} = 1 - R_{01} = 1 - |r_{01}|^2$ $\xrightarrow{\text{双层}} (1 - R_{01})^2$

求解波动方程 $k = \beta + i\alpha$ $\alpha = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{-1 + \sqrt{1 + \left(\frac{\epsilon''}{\epsilon}\right)^2}}$

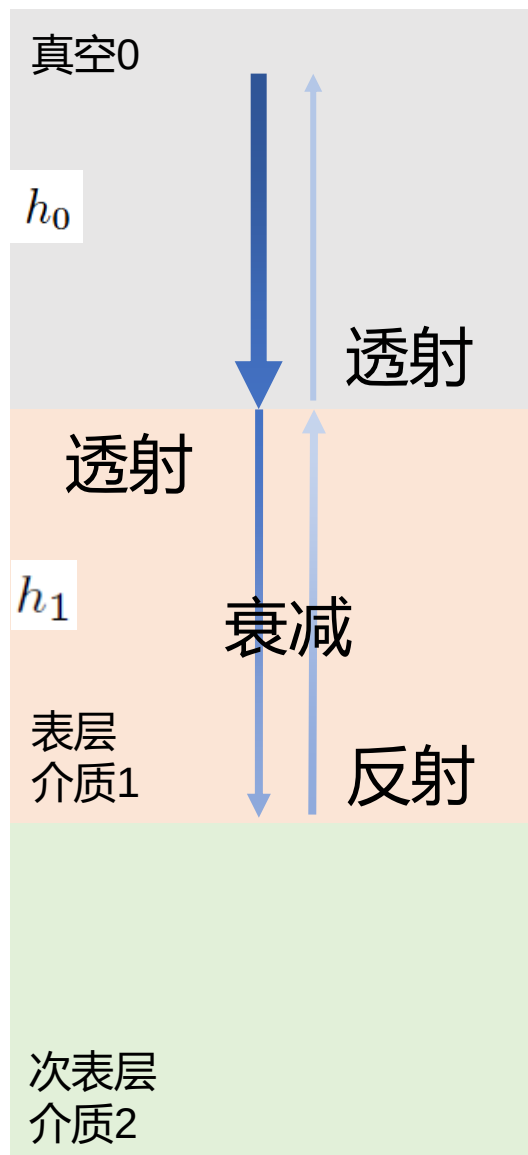
衰减 电场衰减 $e^{-\alpha_1 z}$ 功率衰减 $e^{-2\alpha_1 z}$ $\xrightarrow{\text{往返}} e^{-4\alpha_1 h_1}$

反射 $r_{12} = \frac{\tilde{n}_1 - \tilde{n}_2}{\tilde{n}_1 + \tilde{n}_2}$ $\longrightarrow |r_{12}|^2$

$$P_b = \underbrace{\left(P_t \frac{G^2 \lambda^2}{64\pi^2 (h_0 + h_1)^2}\right)}_{\text{系统与几何项}} \times \underbrace{(1 - R_{01})^2}_{\text{双程表面透射}} \times \underbrace{e^{-4\alpha_1 h_1}}_{\text{双程介质衰减}} \times \underbrace{R_{12}}_{\text{基底反射}}$$

$$= P_t \frac{G^2 \lambda^2}{64\pi^2 (h_0 + h_1)^2} \left[1 - \left(\frac{1 - n_1}{1 + n_1}\right)^2\right]^2 \left|\frac{\tilde{n}_1 - \tilde{n}_2}{\tilde{n}_1 + \tilde{n}_2}\right|^2 \exp\left(-4h_1 \omega \sqrt{\frac{\mu\epsilon_1}{2}} \sqrt{-1 + \sqrt{1 + \left(\frac{\epsilon_1''}{\epsilon_1}\right)^2}}\right)$$

雷达的次表层探测



雷达回波时间, 可得 距离 h_0 h_1

考虑, 很大的 h_1 探测到 很大的 P_b

$$P_b = \underbrace{\left(P_t \frac{G^2 \lambda^2}{64 \pi^2 (h_0 + h_1)^2} \right)}_{\text{系统与几何项}} \times \underbrace{(1 - R_{01})^2}_{\text{双程表面透射}} \times \underbrace{e^{-4\alpha_1 h_1}}_{\text{双程介质衰减}} \times \underbrace{R_{12}}_{\text{基底反射}}$$

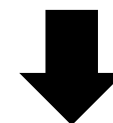
测量-大

测量-小

已知

大?

大?



很小的 α_1 很大的 R_{12}
 纯净的水冰? 液态水?

火星南极地下水的发现



火星南极极冠：干冰，水冰？

火星南极地下水的发现

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REPORT

Radar evidence of subglacial liquid water on Mars

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Orosei et al., 2018, Science

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Multiple subglacial water bodies below the south pole of Mars unveiled by new MARSIS data

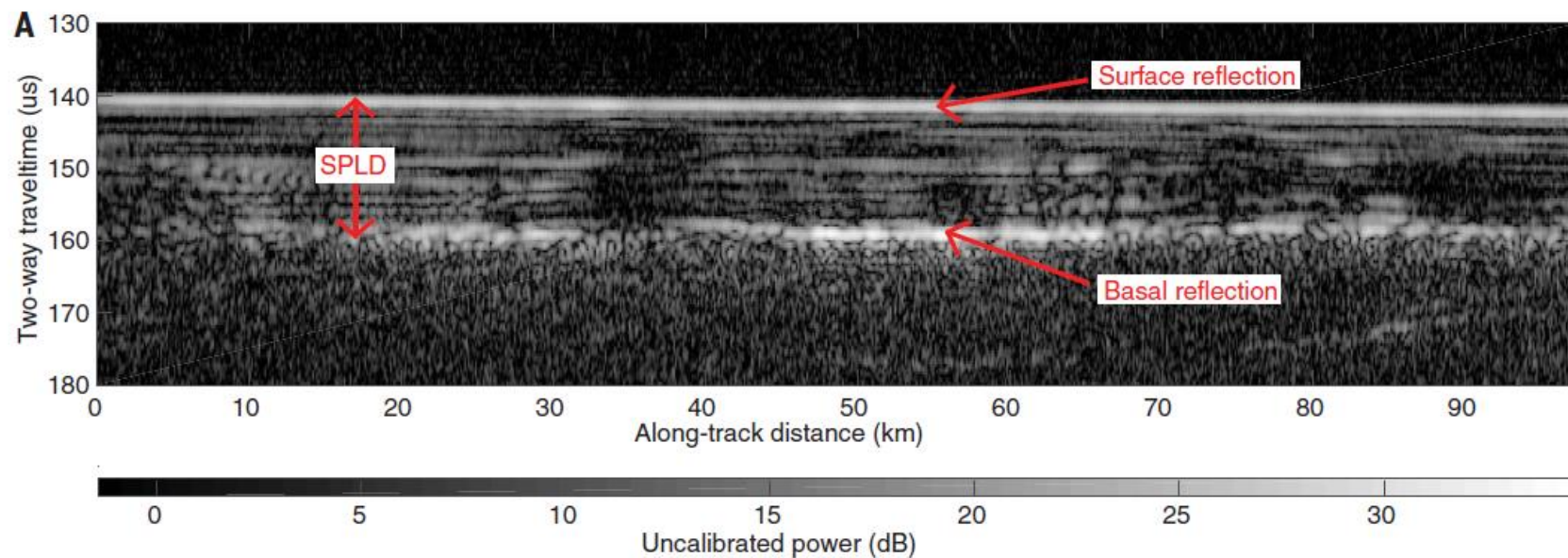
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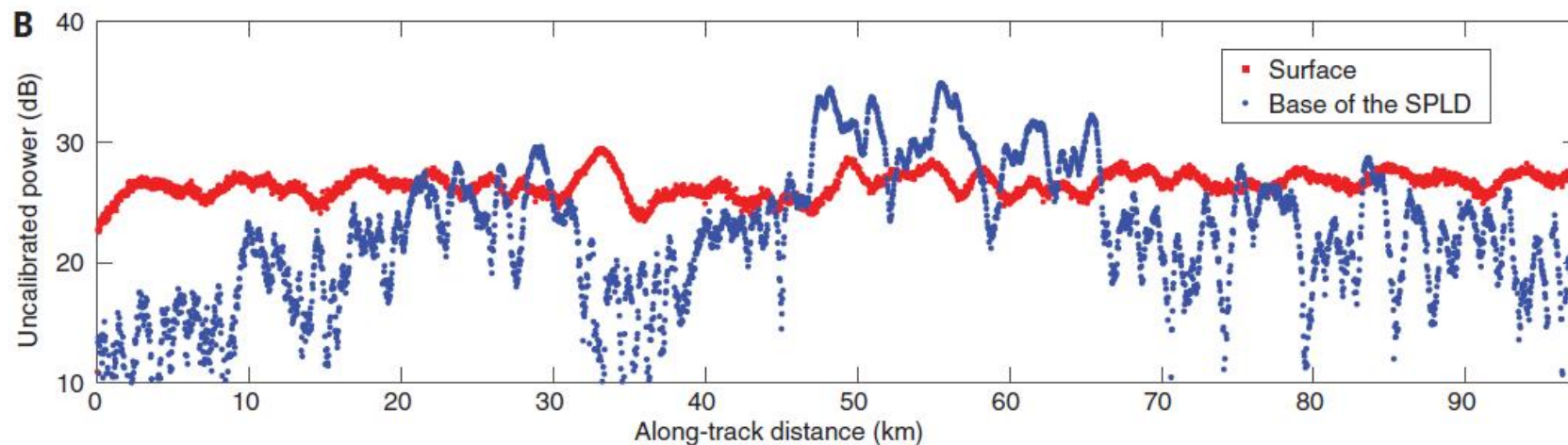
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Lauro et al., 2020, Nature Astronomy

火星南极地下水的发现



明亮——强反射层



反射功率

谢谢!

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2025.12.18

介质的介电常数 影响 电磁波的反射与折射

利用雷达回波, 反演介质的介电常数, 推断介质